43. $(4.5 \times 10^4) \times (1.2 \times 10^6)$
44. $(3.45 \times 10^7) \times (2.81 \times 10^{-3})$
45. $(3.1 \times 10^8) \times (2.7 \times 10^{-5})$
46. $(6.32 \times 10^{-4}) \times (1.31 \times 10^{-5})$
47. $(3.91 \times 10^6) \times (8.13 \times 10^2)$
48. $(7.12 \times 10^{-11}) \times (6.61 \times 10^{-5})$
49. $(3.45 \times 10^4) \div (1.5 \times 10^6)$
50. $(1.4 \times 10^3) \div (5.6 \times 10^{-5})$

For the following exercises, apply your understanding of scientific notation to real-world applications.

51. When stretched out, a strand of human DNA is, on average, 2.066×10^2 cm. One centimeter, or 1 cm, is 1×10^{-5} km. Determine the average length of a strand of human DNA in kilometers.
52. One approximation of the average number of cells in the human body is 3×10^{13} cells (30 trillion!!!). If the DNA of each cell were stretched out and laid end to end, what would be the total length of the DNA in km? Use your answer from Exercise 51 for the length, in kilometers, of DNA.
53. The equatorial circumference of Earth is approximately 4.007×10^4 km. Use the answer from Exercise 52 to determine the number of times that the stretched out human DNA would encircle Earth.
54. The average stride length of a 1.651 m tall woman is 6.6×10^{-1} meters. If such a person could walk from Buffalo, NY, to Buenos Aires, Argentina, in a straight line, how many steps would that person need to take? See Exercise 18 for the distance from Buffalo, NY, to Buenos Aires, Argentina.

3.10 Arithmetic Sequences

Learning Objectives

After completing this section, you should be able to:

1. Identify arithmetic sequences.
2. Find a given term in an arithmetic sequence.
3. Find the nth term of an arithmetic sequence.
4. Find the sum of a finite arithmetic sequence.
5. Use arithmetic sequences to solve real- world applications

As we saw in the previous section, we are adding about 2.5 quintillion bytes of data per day to the Internet. If there are 550 quintillion bytes of data today, then there will be 552.5 quintillion bytes tomorrow, and 555 quintillion bytes in 2 days. This is an example of an arithmetic sequence. There are many situations where this concept of fixed increases comes into play, such as raises or table arrangements.

Identifying Arithmetic Sequences

A **sequence** of numbers is just that, a list of numbers in order. It can be a short list, such as the number of points earned on each assignment in a class, such as {10, 10, 8, 9, 10, 6, 10}. Or it can be a longer list, even infinitely long, such as the list of prime numbers. For example, here's a sequence of numbers, specifically, the squares of the first 12 natural numbers.

{1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144}

Each value in the sequence is called a **term**. Terms in the list are often referred to by their location in the sequence, as in the nth term. For the sequence {1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144}, the first term of the sequence is 1, the fourth term is 16, and so on. In the sequence of assignment scores {10, 10, 8, 9, 10, 6, 10}, the first term is 10 and the third term is 8 (Figure 3.47).

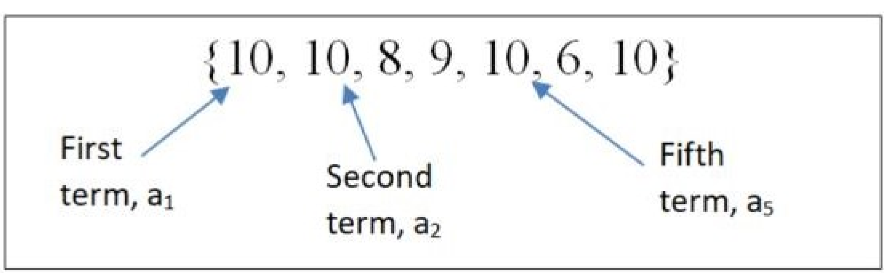

Figure 3.47 Sequence showing first, second, and fifth terms

The notation we use with sequences is a letter, which represents a term in the sequence, and a subscript, which indicates what place the term is in the sequence. For the sequence {1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144}, we will use the letter a as a value in the sequence, and so a_5 would be the term in the sequence at the fifth position. That number is 25, so we can write $a_5 = 25$.

In this section, we focus on a special kind of sequence, one referred to as an **arithmetic sequence**. Arithmetic sequences have terms that increase by a fixed number or decrease by a fixed number, called the **constant difference** (denoted by d), provided that value is not 0. This means the next term is always the previous term plus or minus a specified, constant value. Another way to say this is that the difference between any consecutive terms of the sequence is always the same value.

To see a constant difference, look at the following sequence: {7, 15, 23, 31, 39, 47, 55, 63, 71, 79, 87}. Figure 3.48 illustrates that each term of the sequence is the previous term plus 8. Eight is the constant difference here.

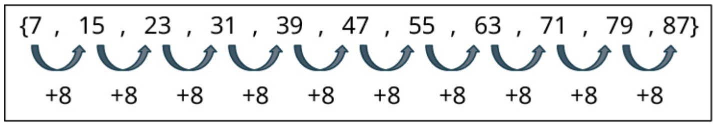

Figure 3.48 Sequence of numbers with 8 added to each term

EXAMPLE 3.132

Identifying Arithmetic Sequences

Determine if the following sequences are arithmetic sequences. Explain your reasoning.

1. $\{4, 7, 10, 13, 16, 19, 22, 25, \ldots\}$
2. $\{20, 40, 80, 160, 320, 640\}$
3. $\{7, 1, -5, -11, -17, -23, -29, -34, -40\}$

Solution

1. In the sequence $\{4, 7, 10, 13, 16, 19, 22, 25, \ldots\}$, every term is the previous term plus 3. The ellipsis indicates that the pattern continues, which means keep adding 3 to the previous term to get the new term. Therefore, this is an infinite arithmetic sequence.
2. In the sequence $\{20, 40, 80, 160, 320, 640\}$, terms increase by various amounts, for instance from term 1 to term 2, the sequence increases by 20, but from term 2 to term 3 the sequence increases by 40. So, this is not an arithmetic sequence.
3. In the sequence $\{7, 1, -5, -11, -17, -23, -29, -34, -40\}$, every term is the previous term minus 6, so this is an arithmetic sequence.

YOUR TURN 3.132

Determine if the following sequences are arithmetic sequences. Explain your reasoning.

1. $\{7.6, 5.4, 3.2, 1.0, -1.2, -3.4, -5.6, -7.8, -10.0\}$
2. $\{14, 16, 18, 22, 28, 40, 32, 0\}$
3. $\{14, 20, 26, 32, 38, 44, 50, 56, 62, 68, 74, 80, \ldots\}$

Arithmetic sequences can be expressed with a formula. When we know the first term of an arithmetic sequence, which we label a_1, and we know the constant difference, which is denoted d, we can find any other term of the arithmetic sequence. The formula for the ith term of an arithmetic sequence is $a_i = a_1 + d \times (i - 1)$.

FORMULA

If we have an arithmetic sequence with first term a_1 and constant difference d, then the ith term of the arithmetic sequence is $a_i = a_1 + d \times (i - 1)$.

Let's examine the formula with this arithmetic sequence: $\{4, 7, 10, 13, 16, 19, 22, 25, \ldots\}$. In this sequence $a_1 = 4$ and $d = 3$. The table below shows the values calculated.

i, Place in Sequence	a_i, i^{th} Term	Value of Term	Term Written as $a_1 + 3 \times (i-1)$
1	a_1	4	$4 + 3 \times 0$
2	a_2	7	$4 + 3 \times 1$
3	a_3	10	$4 + 3 \times 2$
4	a_4	13	$4 + 3 \times 3$
5	a_5	16	$4 + 3 \times 4$
i	a_i		$4 + 3 \times (i-1)$

We can see how the ith term can be directly calculated. In this sequence, the formula is $a_1 + 3 \times (i-1)$ where the first term, a_1, is 4 and the constant difference d is 3. We can then determine the 47th term of this sequence: $a_{47} = 4 + 3 \times (47-1) = 4 + 3 \times 46 = 4 + 138 = 142$.

EXAMPLE 3.133

Calculating a Term in an Arithmetic Sequence

Identify a_1 and d for the following arithmetic sequence. Use this information to determine the 60th term.

$$\{18, 31, 44, 57, 70, 83, ...\}$$

Solution

Inspecting the sequence shows that $a_1 = 18$ and $d = 13$. We use those values in the formula, with $i = 60$.

$$a_{60} = a_1 + d \times (i-1) = 18 + 13 \times (60-1) = 18 + 13 \times 59 = 18 + 767 = 785$$

YOUR TURN 3.133

1. Identify a_1 and d for the following arithmetic sequence. Use this information to determine the 86th term.

   $\{4.5, 8.1, 11.7, 15.3, 18.9, 22.5, 26.1, \ldots\}$

VIDEO

Arithmetic Sequences (https://openstax.org/r/Arithmetic_Sequences)

If we know two terms of the sequence, it is possible to determine the general form of an arithmetic sequence, $a_i = a_1 + d \times (i-1)$.

FORMULA

If we have the ith term of an arithmetic sequence, a_i, and the jth term of the sequence, a_j, then the constant difference is $d = \frac{a_j - a_i}{j-i}$ and the first term of the sequence is $a_1 = a_i - d(i-1)$.

EXAMPLE 3.134

Determining First Term and Constant Difference Using Two Terms

A sequence is known to be arithmetic. Two of its terms are $a_7 = 56$ and $a_{19} = 104$. Use that information to find the constant difference, the first term, and then the 50th term of the sequence.

Solution

To find the constant difference, use $d = \frac{a_j - a_i}{j-i}$. The location of the terms is given by the subscript of the two a terms, $i = 7$ and $j = 19$. So, the constant difference can be calculated as such:

$$d = \frac{104 - 56}{19 - 7} = \frac{48}{12} = 4.$$

The constant difference of 4 is then used to find a_1.

$$a_1 = a_i - d(i-1) = a_7 - 4(7-1) = 56 - 4 \times 6 = 32.$$

So $d = 4$ and $a_1 = 32$.

With this information, the 50th term can be found.

$$a_{50} = a_1 + d \times (i-1) = 32 + 4 \times (50-1) = 32 + 4 \times 49 = 32 + 196 = 228.$$

The 50th term is $a_{50} = 228$.

YOUR TURN 3.134

1. A sequence is known to be arithmetic. Two of the terms are $a_{14} = 41$ and $a_{38} = 161$. Use that information to find the constant difference and the first term. Then determine the 151st term of the sequence.

VIDEO

[Finding the First Term and Constant Difference for an Arithmetic Sequence (https://openstax.org/r/Finding_the_First_Term_and_Constant_Difference_for_an_Arithmetic_Sequence)]

Finding the Sum of a Finite Arithmetic Sequence

Sometimes we want to determine the sum of the numbers of a finite arithmetic sequence. The formula for this is fairly straightforward.

FORMULA

The sum of the first n terms of a finite arithmetic sequence, written s_n, with first and last term a_1 and a_n, respectively, is $s_n = n\left(\frac{a_1 + a_n}{2}\right)$.

EXAMPLE 3.135

Finding the Sum of a Finite Arithmetic Sequence

What is the sum of the first 60 terms of an arithmetic sequence with $a_1 = 4.5$ and $d = 2.5$?

Solution

The formula requires the first and last terms of the sequence. The first term is given, $a_1 = 4.5$. The 60th term is needed. Using the formula $a_1 = a_i + d(i-1)$ provides the value for the 60th term.

$$a_{60} = 4.5 + 2.5(60-1) = 4.5 + 2.5 \times 59 = 4.5 + 147.5 = 152.$$

Applying the formula $s_n = n\left(\frac{a_1 + a_n}{2}\right)$ provides the sum of the first 60 terms.

$$s_{60} = 60\left(\frac{4.5 + 152}{2}\right) = 60 \times \frac{156.5}{2} = 4{,}695.$$

The sum of the first 60 terms is 4,695.

YOUR TURN 3.135

1. What is the sum of the first 101 terms of an arithmetic sequence with $a_1 = 13$ and $d = 2.25$?

VIDEO

Finding the Sum of a Finite Arithmetic Sequence (https://openstax.org/r/Finding_the_Sum_of_a_Finite_Arithmetic_Sequence)

Using Arithmetic Sequences to Solve Real-World Applications

Applications of arithmetic sequences occur any time some quantity increases by a fixed amount at each step. For instance, suppose someone practices chess each week and increases the amount of time they study each week. The first week the person practices for 3 hours, and vows to practice 30 more minutes each week. Since the amount of time practicing increases by a fixed number each week, this would qualify as an arithmetic sequence.

EXAMPLE 3.136

Applying an Arithmetic Sequence

Jordan has just watched *The Queen's Gambit* and decided to hone their skills in chess. To really improve at the game, Jordan decides to practice for 3 hours the first week, and increase their time spent practicing by 30 minutes each week. How many hours will Jordan practice chess in week 20?

Solution

Jordan's practice scheme is an arithmetic sequence, as it increases by a fixed amount each week. The first week there are 3 hours of practice. This means $a_1 = 3$. Jordan increases the time spent practicing by 30 minutes, or half an hour, each week. This means $d = 0.5$. Using those values, and that we want to know the amount of time Jordan will study in week 20, we determine the time in week 20 using $a_i = a_1 + d \times (i - 1)$.

$$a_{20} = 3 + 0.5 \times (20 - 1) = 3 + 0.5 \times 19 = 3 + 9.5 = 12.5$$

So, Jordan will practice 12.5 hours in week 20.

YOUR TURN 3.136

1. Christina decides to save money for after graduation. Christina starts by setting aside \$10. Each week, Christina increases the amount she saves by \$5. How much money will Christina save in week 52?

EXAMPLE 3.137

Finding the Sum of a Finite Arithmetic Sequence

Let's check back in on Jordan. Recall, Jordan had just watched *The Queen's Gambit* and decided to hone their skills, practicing for 3 hours the first week, and increasing the time spent practicing by 30 minutes each week. How many hours total will Jordan have practiced chess after 30 weeks of practice?

Solution

To calculate the total amount of time that Jordan practiced, we need to use $s_n = n\left(\frac{a_1 + a_n}{2}\right)$. The formula requires the first and last terms of the sequence. Since Jordan practiced 3 hours in the first week, the first term is $a_1 = 3$. Because we want the total practice time after 30 weeks, we need the 30th term. Because the constant difference is $d = 0.5$, the 30th

term is $a_{30} = 3 + 0.5(30 - 1) = 3 + 0.5 \times 29 = 3 + 14.5 = 17.5$.

Applying the formula $s_n = n\left(\frac{a_1+a_n}{2}\right)$ provides the sum of the first 30 terms.

$$s_{30} = 30\left(\frac{3 + 17.5}{2}\right) = 60 \times \frac{20.5}{2} = 615.$$

This means that Jordan practiced a total of 615 hours after 30 weeks.

YOUR TURN 3.137

1. In a theater, the first row has 24 seats. Each row after that has 2 more seats. How many total seats are there if there are 40 rows of seat in the theater?

WHO KNEW?

The Fibonacci Sequence

Not all sequences are arithmetic. One special sequence is the **Fibonacci sequence**, which is the sequence that has as its first two terms 1 and 1. Every term thereafter is the sum of the previous two terms. The first nine terms of the Fibonacci sequence are 1, 1, 2, 3, 5, 8, 13, 21, and 34.

This sequence is found in nature, architecture, and even music! In nature, the Fibonacci sequence describes the spirals of sunflower seeds, certain galaxy spirals, and flower petals. In music, the band Tool used the Fibonacci sequence in the song “Lateralus.” The Fibonacci sequence even relates to architecture, as it is closely related to the golden ratio.

VIDEO

Fibonacci Sequence and “Lateralus” (https://openstax.org/r/Fibonacci_Sequence_and_Lateralus)

Check Your Understanding

52. Is the following an arithmetic sequence? Explain.
    {3, 6, 9, 15, 25, 39, 90}
53. What is the 7th term of the following sequence?
    {1, 5, 7, 100, 4, -17, 8, 100, 19, 7.6, 345}
54. In an arithmetic sequence, the first term is 10 and the constant difference is 4.5. What is the 135th term?
55. If the eighth term of an arithmetic sequence is 35 and the 40th term is 131, what is the constant difference and the first term of the sequence?
56. What is the sum of the first 100 terms of the arithmetic sequence with first term 4 and constant difference 7?
57. A new marketing firm began with 30 people in its survey group. The firm adds 4 people per day. How many people will be in their survey group after 100 days?

SECTION 3.10 EXERCISES

For the following exercises, determine if the sequence is an arithmetic sequence.

1. $\{3, 7, 11, 15, 25, 100, \ldots\}$
2. $\{27, 24, 21, 18, 15, 12, 9, \ldots\}$
3. $\{6, -1, -8, -15, -23, -31, -39, \ldots\}$
4. $\{-5, 4, 13, 22, 31, 40, 49, 58, 67, \ldots\}$
5. $\{14, 19, 24, 29, 34, 50, 60\}$
6. $\{3.9, 2.3, 0.7, -0.9, -2.5, -4.1, -5.7, \ldots\}$

7. $\{4, -8, 12, -16, 20, -24, 28, -32, \ldots\}$
8. $\{1, 2, 3, 5, 8, 13, 21, 34, 55, \ldots\}$

For the following exercises, the sequences given are arithmetic sequences. Determine the constant difference for each sequence. Verify that each term is the previous term plus the constant difference.

9. $\{18, 68, 118, 168, 218, 268, \ldots\}$
10. $\{13, 35, 57, 79, 101, 123, 145, 167, \ldots\}$
11. $\{14, 11, 8, 5, 2, -1, -4, \ldots\}$
12. $\{4.5, 1.9, -0.7, -3.3, -5.9, \ldots\}$
13. $\{-27, -13, 1, 15, 29, 43, 57, 71, \ldots\}$
14. $\{3.8, 10.6, 17.4, 24.2, 31, 37.8, 44.6, \ldots\}$

For the following exercises, the first term and the constant difference of an arithmetic sequence is given. Using that information, determine the indicated term of the sequence.

15. $a_1 = 12$, $d = 11$, find a_{20} .
16. $b_1 = 5$, $d = 8$, find b_{38} .
17. $c_1 = 48$, $d = -7$, find c_{50} .
18. $a_1 = 110$, $d = -16$, find a_{27} .
19. $t_1 = 15.3$, $d = 4.2$, find t_{17} .
20. $b_1 = 23.8$, $d = 11.7$, find b_{120} .
21. $b_1 = 27.45$, $d = -3.67$, find b_{40}.
22. $a_1 = 67.4$, $d = -12.3$, find a_{200} .

For the following exercises, two terms of an arithmetic sequence are given. Using that information, identify the first term and the constant difference.

23. $a_5 = 27$, $a_{15} = 77$
24. $b_{10} = 47$, $b_{25} = 137$
25. $a_9 = 38$, $a_{45} = 189.2$
26. $a_6 = 43$, $a_{41} = -377$
27. $a_4 = -12.3$, $a_{54} = -106.5$
28. $a_{12} = 45.9$, $a_{60} = -563.7$

For the following exercises, the first term and the constant difference is given for an arithmetic sequence. Use that information to find the sum of the first n terms of the sequence, s_n.

29. $a_1 = 15$, $d = 7$, calculate s_{10}
30. $a_1 = 2$, $d = 13$, calculate s_{20} .
31. $a_1 = 105$, $d = 0.3$, calculate s_{15} .
32. $a_1 = 56.2$, $d = 1.1$, calculate a_{35} .
33. $a_1 = 450$, $d = -20$, calculate s_{20}.
34. $a_1 = 1400$, $d = -35$, calculate s_{40} .

For the following exercises, apply your knowledge of arithmetic sequences to these real-world scenarios.

35. A collection is taken up to support a family in need. The initial amount in the collection is $135. Everyone places $20 in the collection. When the 35th person puts their $20 in the collection, how much is present in the collection?
36. There are 50 songs on a playlist. Every minute, 3 more songs are added to the playlist. How many songs are on the playlist after 40 minutes have passed?
37. One genre on Netflix has 1,000 shows. Every week, 20 shows are added to that genre. After 15 weeks, how many shows are in that genre?
38. A new local band has 10 people come to their first show. News of the band spreads afterwards. Each week, 4 more people attend their show than the previous week. After 50 weeks, how many people are at their show?
39. The Jester Comic book store is going out of business and is taking in no new inventory. Its inventory is currently 13,563 titles. Each day after, they sell or give away 250 titles. After 15 days, how many titles are left?
40. Jasmyn has decided to train for a marathon. In week one, Jasmyn runs 5 miles. Each week, Jasmyn increased the running distance by 2 miles. How many miles will Jasmyn run in week 13 of the training schedule?
41. A 42-gallon bathtub sits with 14 gallons in it. The faucet is turned on and is now being filled at the rate of 2.2 gallons per minute, but is draining slowly, at 1.8 gallons per minute. After 20 minutes, how many gallons are in the tub?
42. A trained diver is 250 feet deep. The diver is nearly out of air and needs to surface. However, the diver can only comfortably ascend 30 feet per minute. How deep is the diver after ascending for 5 minutes?